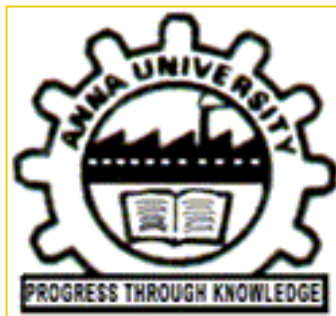


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EC23301 ELECTROMAGNETIC FIELDS

[PROJECT REPORT]

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-NOVEMBER 2024.

Date:

Signature of the Professor



WIRELESS

AC SIGNAL TESTER

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Introduction

Project Overview

The Wireless AC Signal Tester project aims to design and develop a non-contact device capable of detecting the presence of alternating current (AC) voltage in electrical circuits. Unlike traditional testers that require physical contact with the conductor, this wireless device leverages electromagnetic field detection, making it safer and more efficient for identifying live wires and troubleshooting electrical systems.

Need for the Project

Electrical maintenance and troubleshooting often involve risks, especially when working with live wires. Accidental contact with energized circuits can lead to severe injuries or equipment damage. A wireless AC signal tester mitigates these risks by providing a safe and convenient way to detect voltage without direct interaction with the electrical system. Additionally, such a tool is invaluable for quickly identifying faults in circuits and ensuring safety during installations or repairs.

Components Required:

QUANTITY	PART DESCRIPTION
3	Transistors BC547 (NPN)
1	Resistor 100K Ω
1	Resistor 10K Ω
1	Resistor 100 Ω
1	LED (Red)
1	Buzzer 9V
1	9V Battery
1	Battery Snap
1	Bread Board
As required	Wires

Hardware Description:

Transistor:

The BC547 is an NPN transistor used for low-power switching and amplification. Key specs: max voltage 45V, max current 100mA, and power dissipation 500mW. It has 3 pins: Collector, Base, and Emitter. Commonly used in signal amplification, switching, and oscillator circuits.

Resistors:

1. 100k Ω Resistor:

High resistance, commonly used for pull-up or pull-down configurations in circuits. Useful in voltage dividers and low-current applications.

2. 10k Ω Resistor:

Medium resistance, versatile and frequently used in signal conditioning, filters, and biasing. Common in analog and digital circuits.

3. 100 Ω Resistor:

Low resistance, used for current limiting, protection, or low-impedance loads. Common in LED circuits to control current flow.

LED:

Emits light, used as an indicator. Needs a resistor to limit current (e.g., 330 Ω for 9V). Connect anode (+) to positive, cathode (-) to negative.

Buzzer:

Produces sound, used for alerts. Connect directly to power if it's an active buzzer (built-in oscillator). For passive buzzers, a circuit or signal is required to generate sound.

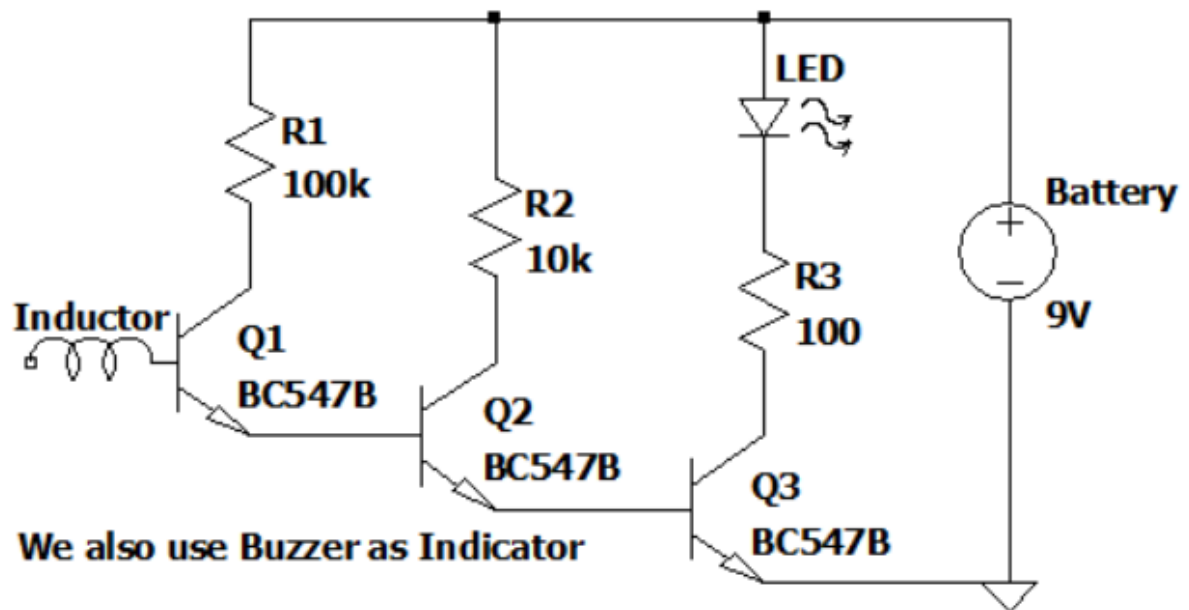
9V Battery:

Type : Commonly rectangular, with a snap connector on top.

Voltage : Nominal voltage of 9V.

Chemistry: Available as Alkaline, Lithium, NiMH (rechargeable), or Zinc-Carbon.

Circuit Diagram:



Working :

The working principle of a Wireless AC Signal Tester is based on fundamental concept in physics: Electromagnetic Induction. These principle allow the device to detect the presence of AC voltage without requiring physical contact with the conductor. Amplifier that is used to amplifies the weak signal induced by the AC field.

Operation

The tester is held close to the conductor.

If the conductor is carrying AC Voltage, the tester detects the electric field or magnetic field.

An indicator (light and sound) is activated to inform the user of the live status.

Electromagnetic Induction

Concept:

When an alternating current (AC) flows through a conductor, it generates a fluctuating electromagnetic field around the wire.

How It Works:

The tester contains a small coil or antenna near its tip, which acts as a sensor.

The fluctuating magnetic field around the live conductor induces a small voltage or current in this sensor coil (according to Faraday's Law of Electromagnetic Induction).

This induced voltage is then amplified and processed by the internal circuitry to detect the presence of AC voltage.

Applications:

1. Detecting live wires in wall outlets or cables.
2. Verifying the presence of voltage in circuit breakers and electrical panels.
3. Identifying broken wires or faults in circuits.

Advantages:

1. Safe and non-contact detection.
2. Portable and easy to use.
3. No interruption to the circuit during testing.

Limitations:

1. May give false positives near strong electromagnetic fields or static charges.
2. Limited sensitivity to low-voltage AC signals.
3. Cannot measure voltage or current levels, only detects their presence.

Future Improvements:

1. Dual Voltage Detection:

Integrating the ability to detect both AC and DC voltages.

2. Improved Sensitivity:

Enhancing the design to detect lower voltages (e.g., <50V).

3. Wireless Connectivity:

Adding Bluetooth or Wi-Fi for real-time monitoring via a smartphone app.

Conclusion:

Wireless AC Signal Testers are indispensable tools in modern electrical work, offering safe and efficient non-contact detection of AC voltage. While they have limitations, ongoing advancements in sensitivity, dual-voltage capabilities, and wireless integration promise to make these devices even more versatile and reliable.